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Impact of time varying interaction: Formation and annihilation of extreme events in dynamical systems

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ABSTRACT

This study investigates the emergence of extreme events in two different coupled systems: the FitzHugh–Nagumo neuron model and the forced Liénard system, both based on time-varying interactions. The time-varying coupling function between the systems determines the duration and frequency of their interaction. Extreme events in the coupled system arise as a result of the influence of time-varying interactions within various parameter regions. We specifically focus on elucidating how the transition point between extreme events and regular events shifts in response to the duration of interaction time between the systems. By selecting the appropriate interaction time, we can effectively mitigate extreme events, which is highly advantageous for controlling undesired fluctuations in engineering applications. Furthermore, we extend our investigation to networks of oscillators, where the interactions among network elements are also time dependent. The proposed approach for coupled systems holds wide applicability to oscillator networks.

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The emergence of extreme events (EE) in coupled oscillators is an important phenomenon that has attracted considerable attention in complex systems. The study of extreme events in coupled oscillators has practical implications in diverse fields, such as climate science, power grids, and neuroscience, where understanding and predicting the occurrence of such events is crucial for system stability and risk assessment. Here, we are investigating the shifting of transition points between extreme events and regular events using time-varying interactions. We start our analysis with two different examples of coupled systems and, finally, extend it to a network of oscillators. We discover the intermittent occurrence of unforeseen extreme events with significantly large amplitudes within a broader parameter space. Intriguingly, our study provides evidence that an appropriate selection of duty cycle percentages and interaction times allows us to exert control over these undesired extreme events according to specific

requirements. This finding emphasizes the importance of understanding the underlying dynamics and employing suitable modulation strategies to effectively manage and mitigate the occurrence of such extreme events with substantial amplitudes.

I. INTRODUCTION

In recent decades, understanding of time-varying interaction (TVI) between distinct coupled oscillators and networks received significant attention, which was used to explore various intricate dynamics in a wide range of complex systems.^{1–3} There are a vast number of studies reported that the interactions among different units or nodes are invariant over time. However, in real situations, many of the interactions between distinct models are either of an adaptive nature or the combination of active and rest states

reveal the existence of time-varying interaction in the systems. Examples are neuronal signal transitions,⁴ gene interaction,^{5,6} social communications,⁷ and epidemic spreading,⁸ to name a few.

In particular, the transition of chemical molecules and electrical signals between neurons is not active for all time and their interaction varies among active and rest states, which are more essential to understanding discrete neuronal communications.⁴ The time-varying structures of various gene regulations are addressed in detail using auto-regressive time-varying models.⁶ In social communications, the social relationship or interaction between a pair can also vary as a function of time.⁷ The TVI induced quenching and revival of oscillation in coupled systems have been explored in Refs. 9–11 and its applications are involved in various disciplines. Moreover, the impact of nonstationary winds, such as downbursts and typhoons, explored using the time-varying extreme value distribution method.¹²

On the other hand, the emergence of extreme events (EE) has been reported in a distinct complex system of laser models,^{13–18} mechanical systems,^{19,20} electronic experiments,^{21,22} coupled maps,^{23–25} neuron models,^{26–28} and others.^{29–33} The FitzHugh–Nagumo (FHN) units of two coupled and different network topologies manifest extreme events for a specific range of system parameters.^{34,35} Moreover, the appearance of EE that is connected with riddle basin boundaries is illustrated in a delay-coupled FitzHugh–Nagumo system.³⁶ Various possibilities to anticipate extreme desynchronize events and their possible control analogies using anticipate synchronization have been reported in a delay-coupled circuit model.³⁷ The prime goal of aforesaid works is to establish a generic mechanism for the formation of EE that would help to further improve the available forecasting metrics and tools. In this line, the distinct sudden transition to extreme events from a bounded state exemplified owing to the dynamical process of interior-crisis,^{23,26,28,29} stick-slip dynamics,¹⁹ multistability induced attractor-hopping,¹⁷ attractor bubbling,³⁸ and distinct forms of intermittencies, such as Pomeau–Manneville intermittency,^{13,39} quasiperiodic intermittency,¹⁶ and on-off intermittency²⁴ in different model systems.

This study focuses on examining EE in the presence of TVI. In existing literature, TVI is alternatively referred to as occasional (or intermittent) interaction, encompassing both deterministic and stochastic variations. For example, in the stochastic on-off coupling scheme,⁴⁰ the activation or deactivation of the coupling term occurs randomly, while in the on-off coupling scheme,⁴¹ the interaction follows a deterministic pattern. Along with the on-off coupling scheme, an alternative analogous deterministic TVI method, known as the transient uncoupling scheme,^{42,43} has also been employed in the literature to investigate various nonlinear phenomena.^{44–47} Notably, in contrast to the on-off coupling method, the transient uncoupling scheme explicitly relies on the phase space coordinates of the interacting oscillators to dictate the time-varying interactions. Numerous other examples of time-varying interacting schemes can be found in the literature, and Ghosh and Chakraborty⁴⁸ have concisely reviewed such schemes. To this end, in this paper, we elaborate more within the purview of deterministic TVI and some glimpse of stochastic time-varying interaction.

Very recently, a few works have focused on the origin of extreme events based on time-varying interaction. The appearance

of EE exemplifies a time-evolving model of interaction among moving agents⁴⁵ and frustrated networks.⁴⁹ In addition, the time-varying effect induced EE in a coupled system by turning the coupling parameter for both explicit and implicit time depended is reported in Ref. 50. Furthermore, various feedback mechanisms and perturbation strategies have been put forth to address the occurrence of undesired, infrequent occasional large-amplitude events within different nonlinear dynamical systems,^{38,45,51} particularly within engineering applications. Nevertheless, the control of natural extreme events remains elusive. In this context, we have employed a straightforward and dependable method to suppress oscillations, including extreme events, within both coupled oscillators and oscillator networks.

Based on the aforementioned understanding and motivation, in this study, we illustrate the influence of time-varying interaction in the formation of extreme events in two different paradigmatic models: the FitzHugh–Nagumo neuron model and the forced Liénard system, respectively. The emergence of extreme events (EE) has been reported in the FHN model with various configurations, including homogeneous and heterogeneous systems,^{28,52} a ring of coupled oscillators,⁵³ and the role of edges in time-dependent interaction networks.⁵⁴ Additionally, the dynamical origin of EE and their distinct transitions have been reported in different forms of the Liénard system.^{21,51,55–57} However, our study presents the following observations: (i) the role of time-varying interaction in the formation and annihilation of EE in coupled oscillators; (ii) when analyzing the formation of extreme events in a model system, the dynamic or time-varying interaction effect can create unexpected extreme events that do not appear during static interaction between systems, highlighting the importance of considering this issue for deeper insights and exploring substantial possibilities for the occurrence of extreme events in a model system; and (iii) through the appropriate selection of time-varying parameter values, we obtained the possibility of eliminating undesired oscillations, including extreme events, in discrete network configurations.

The following parts of the paper are presented as follows. Section II deals with the origin of EE in the FHN neuron model and their distinct transition with the influence of the time-varying effect. The significant impact in the transition point with the response of distinct time-varying parameters is analyzed in a coupled Liénard system, and the results are illustrated in Sec. III. The emergence and annihilation of extreme events in a discrete network of oscillators explored in Sec. IV. The overall conclusion is given in the final section of the paper.

II. FITZHUGH-NAGUMO MODEL

In this section, we elucidate the influence of time-varying interaction in a coupled FitzHugh–Nagumo model, which is represented as the following differential equations:²⁸

$$\begin{aligned} \dot{x}_1 &= x_1(a - x_1)(x_1 - 1) - y_1 + kH(t)(x_2 - x_1), \\ \dot{y}_1 &= b_1x_1 - cy_1, \\ \dot{x}_2 &= x_2(a - x_2)(x_2 - 1) - y_2 + kH(t)(x_1 - x_2), \\ \dot{y}_2 &= b_2x_2 - cy_2. \end{aligned} \quad (1)$$

Here, we used a step function ($H(t)$) which creates a time-varying interaction between two neurons. The interaction function $H(t)$ exhibits the periodic step function of period T represented as

$$H(t) = \begin{cases} \alpha, & 0 < t \leq \tau, \\ \beta, & \tau < t \leq T. \end{cases} \quad (2)$$

The value of $H(t)$ in this function is determined by the value of t within two intervals: $0 < t \leq \tau$ and $\tau < t \leq T$. If t is greater than 0 and less than or equal to τ , $H(t)$ takes the value of α . If t is greater than τ and less than or equal to T , $H(t)$ takes the value of β . The constants α and β play a crucial role as they represent the deciding factor and strength of interaction between the systems.

For instance, if $\alpha = \beta = 1$, it indicates that both systems are interacting throughout the entire time period. Conversely, if $\alpha = \beta = 0$, it signifies that there is no interaction between the systems. In the case where $\alpha = 1$ and $\beta = 0$, the systems will interact for the duration between 0 and τ , and they will remain uncoupled for the duration between τ and T . Hence, Eq. (2) mathematically defines a square wave signal characterized by alternating amplitudes of 0 and 1, with a time period denoted as T and a duty cycle represented by τ . This particular time-varying coupling scheme is called “on-off coupling”⁵⁸ in the literature. The on-off coupling scheme belongs to the category of occasional coupling schemes, wherein the coupling between oscillators is intermittently activated or deactivated. The concept of occasional coupling, particularly the utilization of on-off coupling, has previously been introduced in the literature to study the enhancement of synchronization^{59,60} and amplitude death^{61,62} regions along the control parameter axis. To this end, we can define the interaction time $\tau' := \tau/T$ as the fraction of T over which the coupling remains activated.¹⁰

The system's parameter values for Eq. (1) are chosen as follows: $a = -0.025794$, $b_1 = 0.0065$, $b_2 = 0.0135$, $c = 0.02$, and k represents the coupling strength between the systems.

When we pause the TVI and gradually increase the coupling strength k , the system demonstrates successive period-doubling routes to chaos, followed by the occurrence of extreme events. Just before reaching $k = 0.128$, the system undergoes interior crises. Prior to the crisis point, the system evolves chaotically, but its behavior is mostly confined to a relatively small region of the state space. Beyond the crisis point, the system continues to exhibit bounded chaotic dynamics for the majority of its evolution. However, it sporadically and rarely escapes from its confined region in the state space, giving rise to extreme events or excursions. Following each excursion, the system returns to its bounded chaotic oscillatory state until the occurrence of the next event. The occurrence of extreme events and interior crises in the system can be attributed to the collision of different bifurcation processes, namely, period doubling and reverse period adding. Previous studies have suggested that the collision of these opposing bifurcation processes, which have distinct topological entropies, can lead to the manifestation of an interior crisis.²⁸

At a critical coupling strength of $k = 0.128$, the system occasionally produces extreme events from its bounded chaotic motion. The temporal dynamics of the system are illustrated in Fig. 1(a). We have also plotted a probability distribution function (PDF) for a significantly long time series data, depicted in Fig. 1(b), which clearly

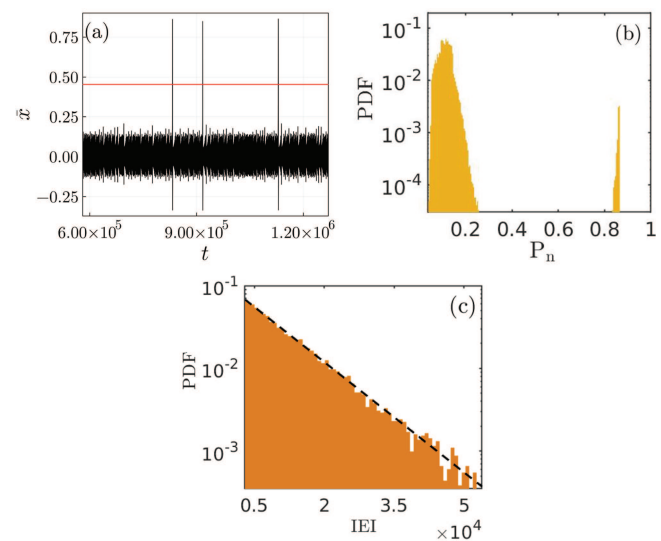


FIG. 1. Coupled FitzHugh–Nagumo model: (a) Time series ($\bar{x} = \frac{x_1+x_2}{2}$) and (b) probability distribution function for the coupling parameter $k = 0.128$ manifest extreme events from the bounded chaotic motion. (c) Probability distribution of IEI, which is fitted by an exponential black dashed line.

indicates a lower probability of extreme events, thus confirming the characteristic nature of extreme events in the FHN model. Further, we have analyzed the inter-event interval (IEI) distribution, i.e., calculate the time interval between the consecutive extreme events for a t-span length of 5×10^9 manifest a Poisson-like distribution in the semi-log scale as shown in Fig. 1(c). The dashed black line in Fig. 1(c) signifies the exponential decay function $P(\text{IEI}) = ae^{(-b\text{IEI})}$ and we have used the fitting parameter values as $a = 0.09236$ and $b = 1.0230 \times 10^{-4}$.

We start by discussing the TVI in the FHN model. In Fig. 2(a), we present a bifurcation diagram for a fixed value of $\tau' = 1.0$ with $T = 1.0$. We set $\alpha = 1$ and $\beta = 0$. Under these conditions, where there is no time-varying interaction between the neurons, the FHN model demonstrates a period-doubling route to chaos followed by a sudden large expansion, as depicted in Fig. 2(a). Interestingly, the system exhibits a small region of extreme events within the range of the coupling parameter $k \in (0.128, 0.12815)$. This is confirmed by plotting the EE qualifier threshold H_s as a horizontal red line. Here, we define $H_s = \langle x_n \rangle + 6\sigma$, where x_n represents the peaks of the time series.

Next, we investigate the influence of time-varying effects in the system by adjusting the value of τ' from 1.0 to 0.8. In Fig. 2(b), we present the bifurcation diagram for a range of coupling strengths $k \in (0.12, 0.16)$. When we modify the interaction time to $\tau' = 0.9$, implying that 90% of the time duration T the neurons interact while 10% of the time they remain independent, the critical parameter k for the occurrence of extreme events shifts to the right side of the bifurcation diagram, specifically around $k \approx 0.132$. This shows that the system experiences a significant shift in the transition point, characterized by a notable expansion occurring at

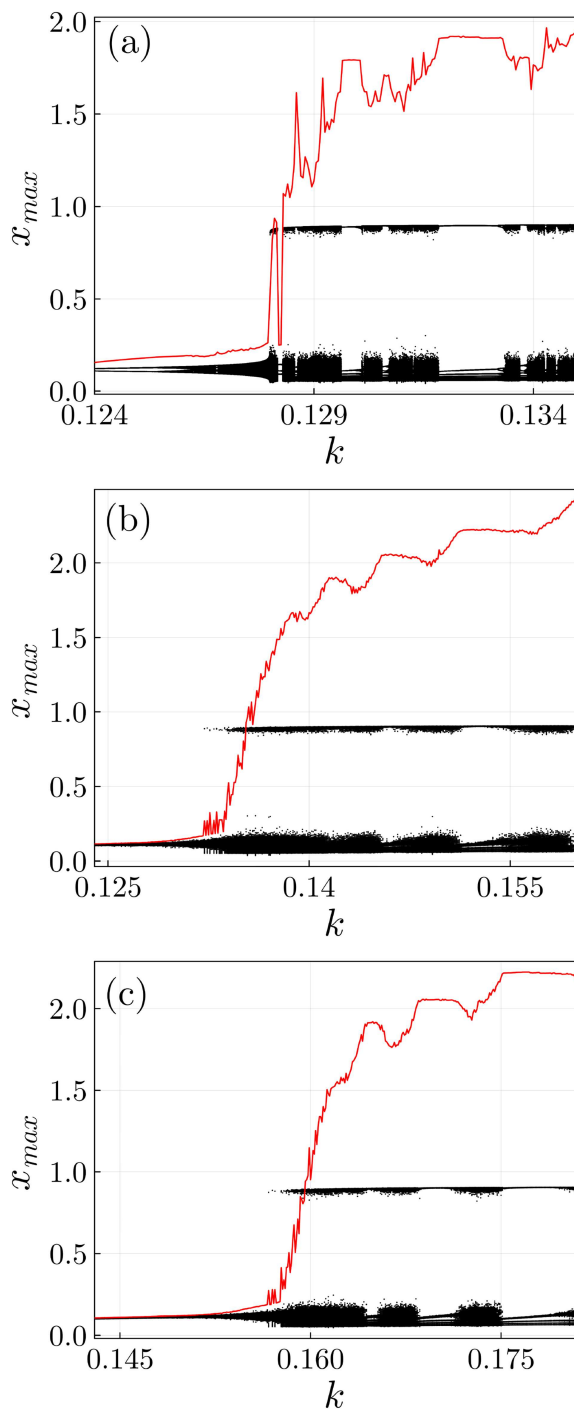


FIG. 2. Bifurcation diagram depicting the influence of varying duty cycles on system behavior: Transition from bounded chaotic states to extreme events under critical coupling parameters k_c . (a) $k_c = 0.128$ with $\tau' = 1.0$, (b) $k_c = 0.132$ with $\tau' = 0.9$, and (c) $k_c = 0.158$ with $\tau' = 0.8$, highlighting the substantial effects of time-varying interactions in the FHN model. For all three cases, the T is fixed at 1.0.

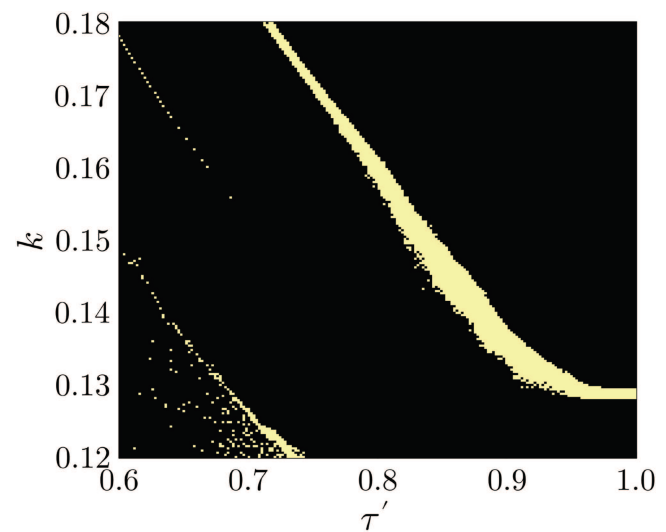


FIG. 3. Phase diagram of the FHN model between the parameters τ' and k . Existence of extreme events (yellow region) and non-extreme events in a wide range of parameters is plotted.

$k \approx 0.132$. Remarkably, in the absence of time-varying interaction, the region of $k \in (0.132, 0.135)$ in Fig. 2(a) corresponds to chaotic or mixed-mode oscillations.²⁸ This change also leads to the absence of extreme events for the parameter $k = 0.128$. Consequently, the time-varying interaction shifts the bifurcation point for the occurrence of extreme events, effectively eliminating the extreme events observed previously with $\tau' = 1$.

Moreover, Fig. 2(c) illustrates the transition of the extreme event region within the range $k \in (0.14, 0.18)$ for $\tau' = 0.8$. This demonstrates the gradual shift in the critical point for the occurrence of extreme events in response to changes in the interaction time. As the interaction time decreases, more coupling strength required for the generation of extreme events increases.

To provide a comprehensive understanding of the occurrence of extreme events across a broader parameter range, we present a phase diagram in Fig. 3, which illustrates the relationship between the τ' and the coupling strength k . In this diagram, the dark gray and yellow regions represent the non-extreme events and extreme events regions, respectively. These regions are distinguished based on the measure of the extreme events qualifier threshold (H_S). From the analysis depicted in Fig. 3, it is evident that the region associated with extreme events gradually shifts as the τ' varies within the range $\tau' \in (0.75, 1.0)$. Conversely, within the range $\tau' \in (0.6, 0.75)$, the occurrence of extreme events appears to be more random.

Similar to the interaction duration between the neuron models, the interaction time T also plays a crucial role in the occurrence of extreme events within specific parameter regions. To investigate how the time period T of the interaction influences the emergence of extreme events, we constructed phase diagrams that distinguish between non-extreme events and extreme events for two different T

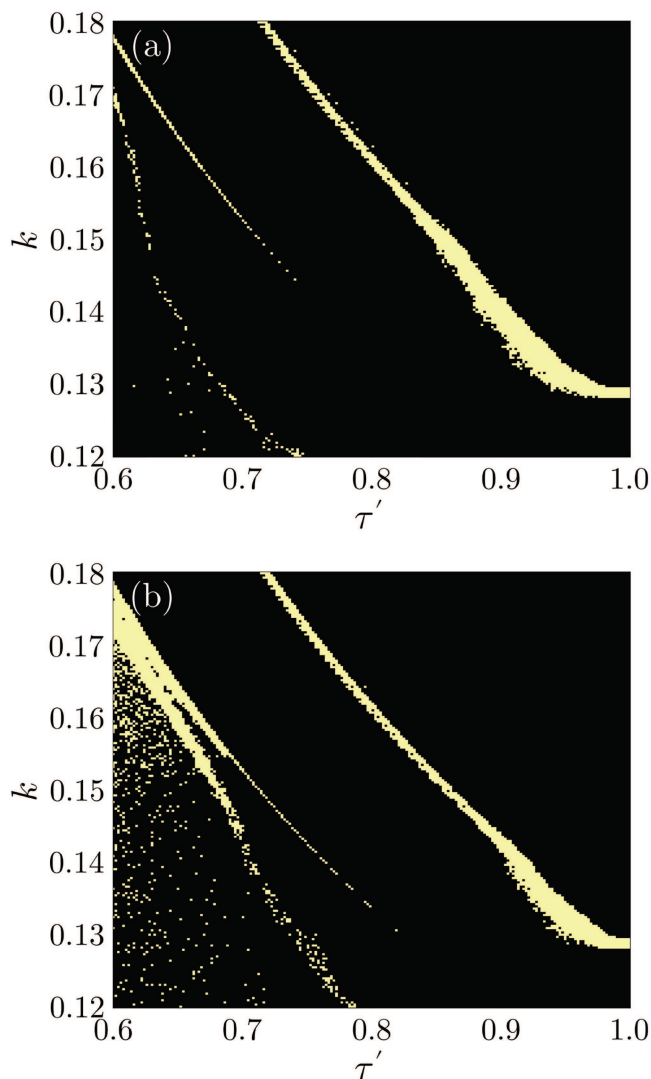


FIG. 4. Phase diagrams of the FitzHugh–Nagumo model: For different values of interaction time (a) $T = 2.0$, and (b) $T = 4.0$, the model proves the appearance of EE in discrete parameter regions.

values such as 2.0 and 4.0. These diagrams are shown in Figs. 4(a) and 4(b), respectively.

Analyzing the two-phase diagrams in Figs. 4(a) and 4(b), we observe a significant alteration in the parameter region associated with EE in the FHN model. This indicates that when addressing the control of extreme events, it is crucial to consider the influence of time-varying interaction (TVI) within the system.

Furthermore, we have observed that by reducing τ' to 0.6 while maintaining a constant interaction time T of 1.0 and varying the coupling strength within the range $k \in (0.13, 0.18)$, the coupled FHN model demonstrates controlled oscillations. Notably, this leads to the complete elimination of extreme events within the system.

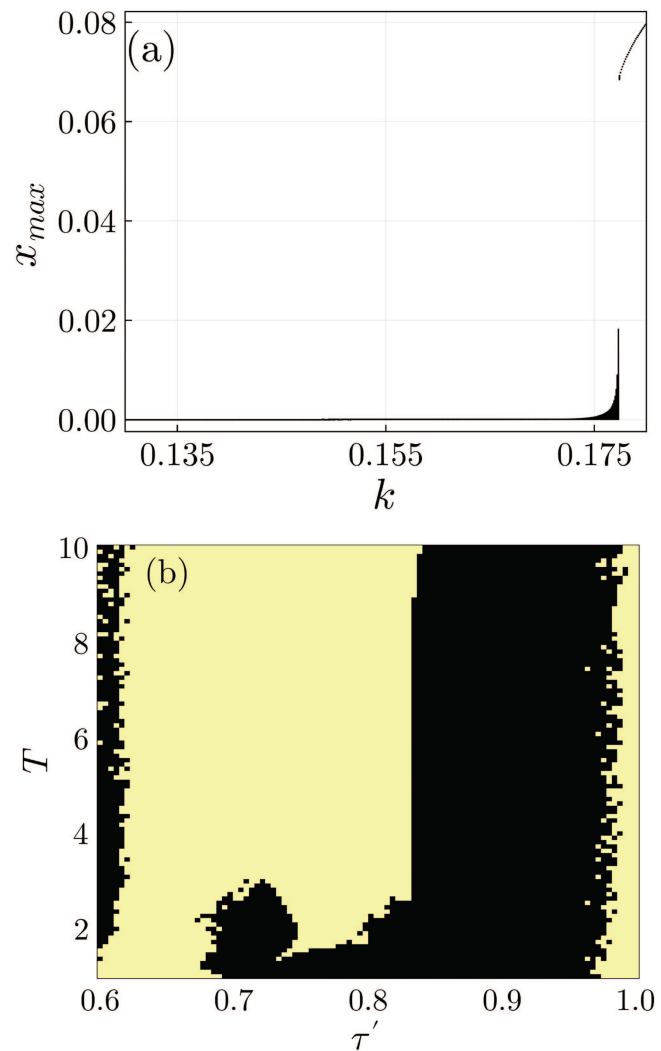


FIG. 5. (a) Bifurcation diagram for the fixed $\tau' = 0.6$ and $T = 1.0$ and (b) two-phase diagram for fixed $k = 0.128$ manifest control of extreme events with response of duty cycle percentage τ' and iteration time T .

The outcome is depicted in Fig. 5(a) through the bifurcation diagram. Additionally, we explore the behavior by plotting a two-phase diagram. This is done with a fixed coupling strength of $k = 0.128$, and varying τ' and T across two parameter phases, as depicted in Fig. 5(b). The visual representation in Fig. 5(b) clearly indicates that appropriate selections of duty cycle percentage and interaction time can effectively eliminate extreme events in specific applications. We also explore the option of introducing stochastic variations between the systems. We examine two specific scenarios: (i) where $\alpha = 1$ and $\beta = 0$, with modulation by a small random fluctuation or periodic oscillation, and (ii) the introduction of a stochastically varying duty cycle. Detailed results for these scenarios can be found in the Appendix.

III. COUPLED LIÉNARD SYSTEM

In order to confirm the generality of the above observation, we have taken another model of the coupled Liénard system²¹ denoted by the following differential equation:

$$\begin{aligned}\dot{x}_1 &= y_1, \\ y_1 &= -\alpha x_1 y_1 - \gamma x_1 - \beta x_1^3 + A \sin(\omega_1 t) + kH(t) \cdot (x_2 - x_1), \\ \dot{x}_2 &= y_2, \\ y_2 &= -\alpha x_2 y_2 - \gamma x_2 - \beta x_2^3 + A \sin(\omega_2 t) + kH(t) \cdot (x_1 - x_2).\end{aligned}\quad (3)$$

Similar to the previous case, the TVI function between the coupled systems is denoted as $H(t)$, while the other system parameters remain fixed: $\alpha = 0.45$, $\beta = 0.5$, $\gamma = -0.5$, $A = 0.2$, $\omega_1 = 0.739$, and $\omega_2 = 0.740$. These parameters represent nonlinear damping or position-dependent damping (α), the strength of nonlinearity (β), the internal frequency of the system γ , and the amplitude (A) and frequencies (ω_1 and ω_2) of the external forcing, respectively. We introduce a small mismatch in the forcing frequency within the system. As we gradually increase the coupling strength k in the absence of the $H(t)$ function, the system undergoes a transition from a bounded chaotic state to a sudden, significant expansion. The transition to EE occurs at a critical coupling value of $k \approx 0.00185$. The temporal dynamics of the extreme events, originating from the bounded chaotic state, are illustrated in Fig. 6(a), and their corresponding probability distribution function (PDF) exhibits a heavy-tail distribution, as depicted in Fig. 6(b). The threshold line (red line: $H_S = \langle y_1 \rangle + 6\sigma$) in Fig. 6(a) confirms the presence of extreme events in the given time series data. The inter-event interval distribution plot for extreme events' time series portrayed in Fig. 6(c), which is fitted by an exponential function $P(\text{IEI}) = ae^{(-b\text{IEI})}$ and its corresponding parameter values are $a = 0.09887$ and $b = 7.114 \times 10^{-6}$.

The impact of TVI in the coupled Liénard system is illustrated by plotting the bifurcation diagram for different values of τ' while keeping $T = 1.0$ fixed. The obtained results are presented in Figs. 7(a)–7(c). In Fig. 7(a), where $\tau' = 1.0$, the system undergoes a transition from chaotic behavior to a sudden, significant expansion at $k \approx 0.00185$. When we change the duty cycle percentage to $\tau' = 0.8$, the point of sudden expansion shifts from $k = 0.00185$ to $k = 0.00218$. Figure 7(b) clearly depicts the alteration in the bifurcation or expansion of the attractor in the coupled Liénard system. Moreover, by decreasing the interaction time to $\tau' = 0.6$, the transition from bounded chaotic motion to an EE occurs at $k \approx 0.00295$, as shown in Fig. 7(c). These changes in the expansion of the attractor with respect to the interaction time reveal the influence of TVI in the system. The results obtained in the Liénard system are consistent with those observed in the coupled FHN neuron model, demonstrating the robustness of the findings. In order to determine the parameter region where EE occurs by varying the duty cycle percentage and the coupling parameter k , we have constructed a phase diagram for a fixed value of $T = 1.0$, as shown in Fig. 8(a). The regions indicating extreme events (yellow) and non-extreme events (dark gray) are differentiated based on the extreme event qualifier threshold. Furthermore, we have examined the influence of the

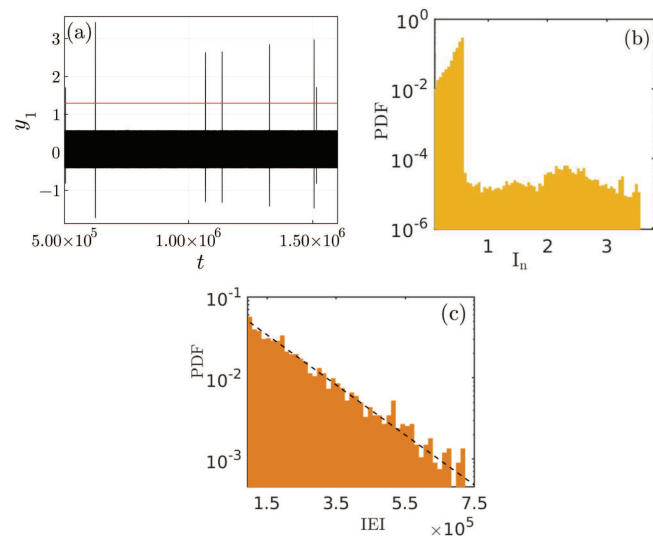


FIG. 6. Coupled Liénard system: (a) time series and (b) probability distribution function for the coupling parameter value $k \approx 0.00185$ manifest a heavy-tail distribution. (c) The interevent interval distribution of extreme events signifies a Poisson-like distribution, which is fitted by an exponential function.

interaction time T in the Liénard system by presenting phase diagrams for $T = 2.0$ and $T = 4.0$, as depicted in Figs. 8(b) and 8(c), respectively. These diagrams illustrate significant changes in the region of the emergence of extreme events within a wider parameter range.

Similar to the previous case, we have shown that for a suitable choice of interaction time T and duty cycle percentage τ' , the extreme large-amplitude events are completely eliminated in the coupled Liénard system. This is illustrated in Fig. 9 by varying the interaction time (T) belonging to (1, 10) and duty cycle percentage (τ') belong to (0.6, 1.0). The other system parameters are fixed as $\alpha = 0.45$, $\beta = 0.5$, $\gamma = -0.5$, $A = 0.2$, $\omega_1 = 0.739$, and $\omega_2 = 0.740$. Moreover, in Sec. IV, we broaden our observation to establish that the effective control over undesired EE, achieved through time-varying interactions in coupled oscillators, extends its validity to oscillator networks as well.

IV. TIME-VARYING INTERACTION EFFECT: NETWORK OF OSCILLATORS

In Secs. I–III, we have discussed the influence of time-varying interactions in the formation of extreme events and the control factors to prevent extreme events (duty cycle percentage, coupling strength, and interaction time). In this section, we extend the study to a different network of oscillators.

First, we elaborate on the TVI effect in an n globally coupled FHN units, which is represented as the following differential

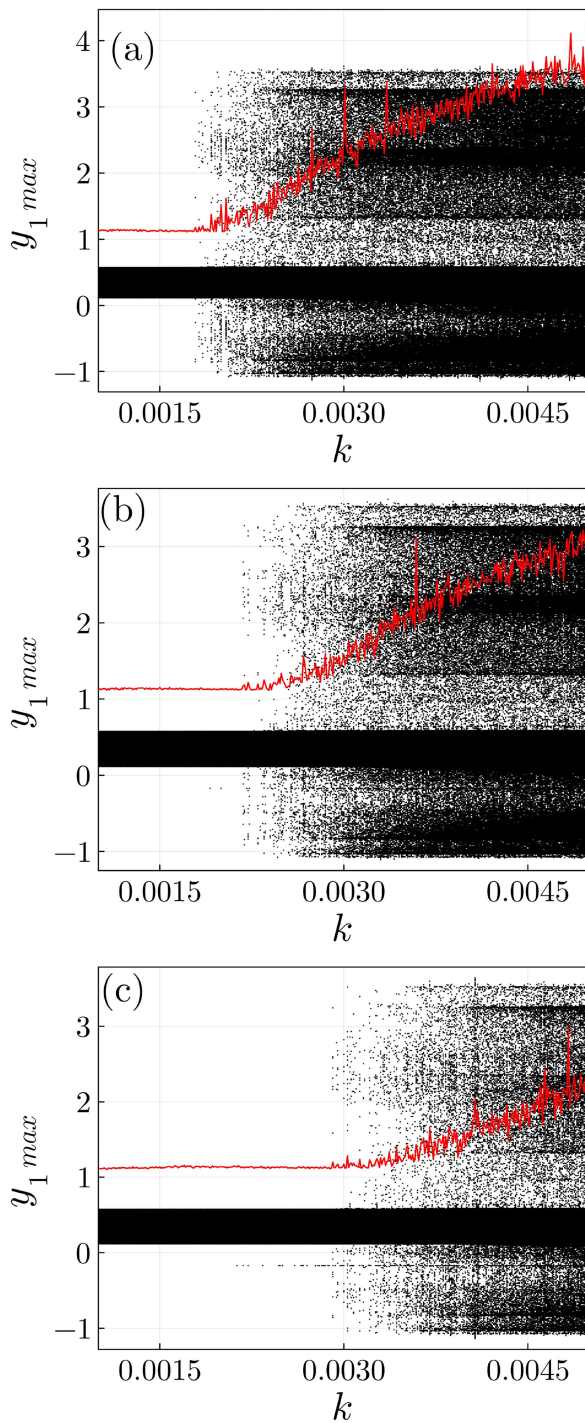


FIG. 7. Bifurcation diagram of the coupled Liénard system manifests shifts in the transition from bounded chaotic motion to sudden surge of systems' amplitude with the influence of time-varying interaction. The transition point of extreme events changing with the influence of TVI: (a) $k = 0.00185$, (b) $k = 0.00218$, and (c) $k = 0.00295$.

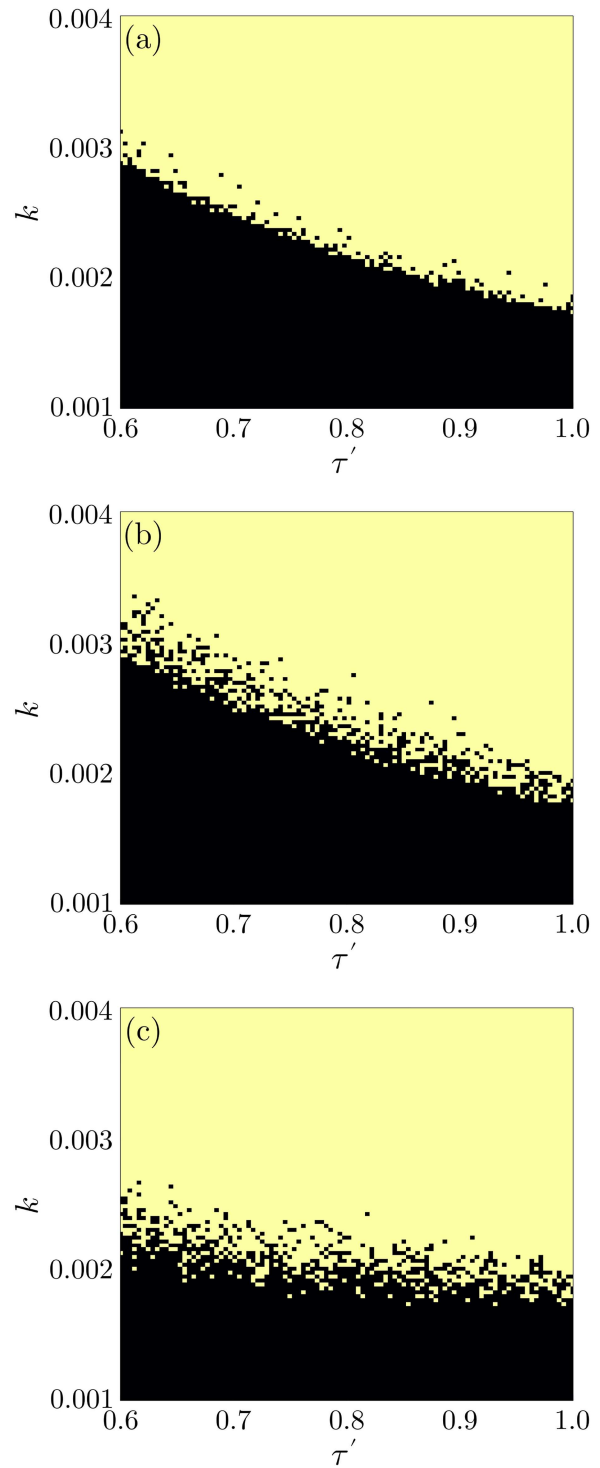


FIG. 8. Phase diagram of coupled Liénard system: Variation in the extreme (yellow) and non-extreme events (black) region with the influence of interaction time $T = 1.0$ (a), $T = 2.0$ (b), and $T = 4.0$ (c), respectively.

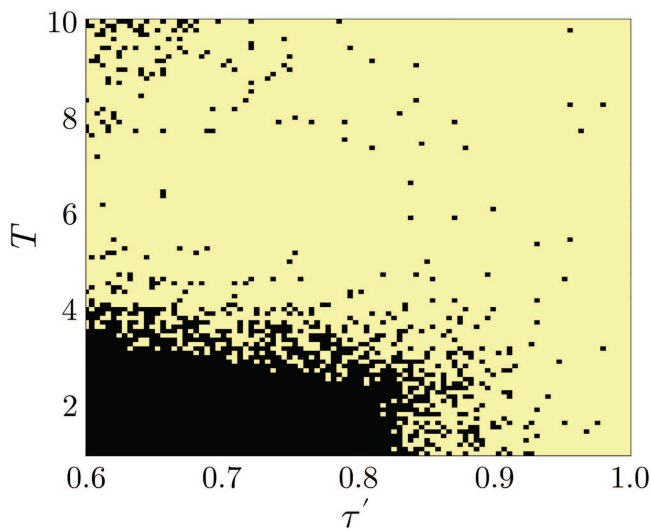


FIG. 9. Phase diagram of coupled Liénard system: emergence of extreme events for the control parameter $k = 0.0021$ is mitigated for the suitable choice of interaction time T and duty cycle percentage τ' .

equation:

$$\begin{aligned}\dot{x}_i &= x_i(a - x_i)(x_i - 1) - y_i + \frac{k}{n} \sum_{j=1}^n A_{ij} H(t)(x_j - x_i), \\ \dot{y}_i &= b_i x_i - c y_i.\end{aligned}\quad (4)$$

To investigate the emergence and mitigation of extreme events in a network of oscillators, we have conducted an analysis using $n = 20$ FHN units. The connectivity matrix, denoted as A_{ij} , defines the topology of the network. The system parameters were set as follows: $a = -0.02651$, $c = 0.02$, $T = 1.0$, $\tau' = 1.0$. Additionally, we introduced mismatched b_i values, evenly distributed in the range of $(0.006, 0.014)$, and $k = 0.145$. Under these conditions, the 20 FHN oscillators exhibited infrequent extreme events, which are visualized in the spatial-temporal plot shown in Fig. 10. Furthermore, we examined the control of EE in the oscillator network, considering the influence of time-varying interaction parameters (τ' and T). The investigation result is depicted in Fig. 11, highlighting the significance of employing a time-varying interaction approach to effectively manage unexpected EE within the complex network.

Furthermore, to provide a clear interpretation of the control of extreme events, we have plotted the temporal evolution of different oscillators: five, seven, and ten. This presentation can be found in Fig. 12 for two scenarios: $T = 1.0$, $\tau' = 1.0$ (a) before control, and $T = 1.0$, $\tau' = 0.5$ (b) after control. Notably, the results depicted in Figs. 12(a) and 12(b) illustrate that the time-varying interactions not only control the undesired extreme events but also offer the capability to suppress the complete oscillatory states within an oscillator network. This is particularly evident in the spatial-temporal plot showcased in Fig. 12(c), which distinctly portrays the suppression of oscillations across all units of the FHN model. It is well

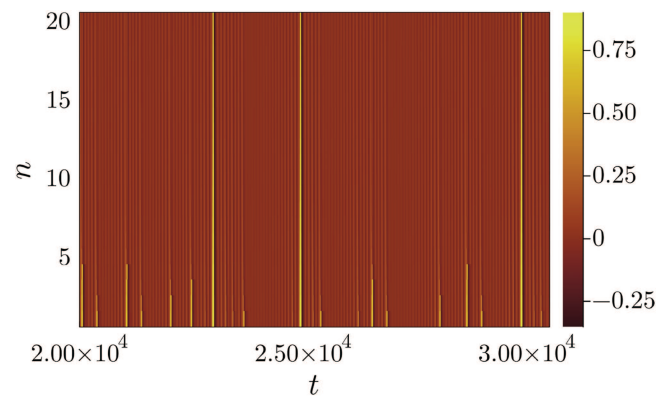


FIG. 10. Temporal evolution globally coupled FHN model: extreme events appeared for the coupling parameter $k = 0.145$, $T = 1.0$, and $\tau' = 1.0$.

established that various time-varying interaction approaches have been utilized to effectively eliminate oscillations in systems, leading to a phenomenon known as amplitude death in coupled oscillators.⁶³ In addition, the appearance of amplitude death dynamics identified in a network of mobile oscillators⁶⁴ and explosive death dynamics explored using dynamic coupling configuration.⁶⁵ However, in this study, we have used a simple and effective approach to eradicate extreme events as well as oscillatory dynamics in a network of FHN units using the suitable choice of time-varying parameters.

In addition, we investigated whether the mitigation of extreme events and quenching of oscillations hold true for a different network configuration. Consequently, we observed similar control of extreme events and amplitude death dynamics in a small-world

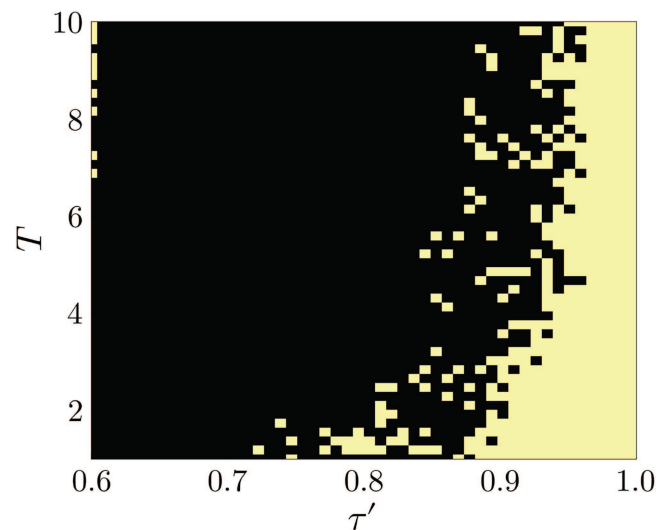


FIG. 11. Phase diagram of a network of 20 globally coupled FHN models. The appearance of rare events for the coupling strength $k = 0.145$ is controlled for the suitable choice of interaction time T and duty cycle percentage τ' .

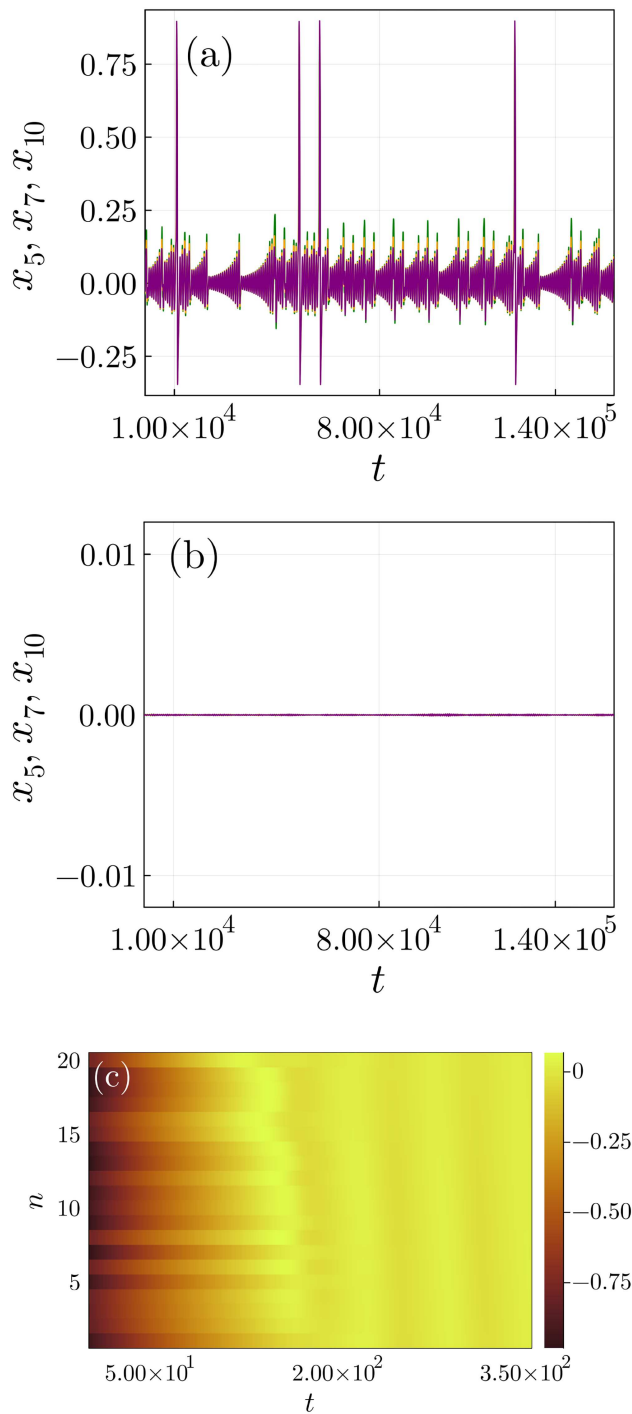


FIG. 12. Quenching of oscillation in a globally coupled network of FHN model: time series of discrete system variables (x_5 , x_7 , and x_{10}) manifest occasional rare events for $\tau' = 1.0$ and $T = 1.0$ (a), whereas the system attains rest state (amplitude death) for $\tau' = 0.5$ and $T = 1.0$ (b). Amplitude death dynamics for a network of oscillator at $T = 1.0$ and $\tau' = 0.5$ (c).

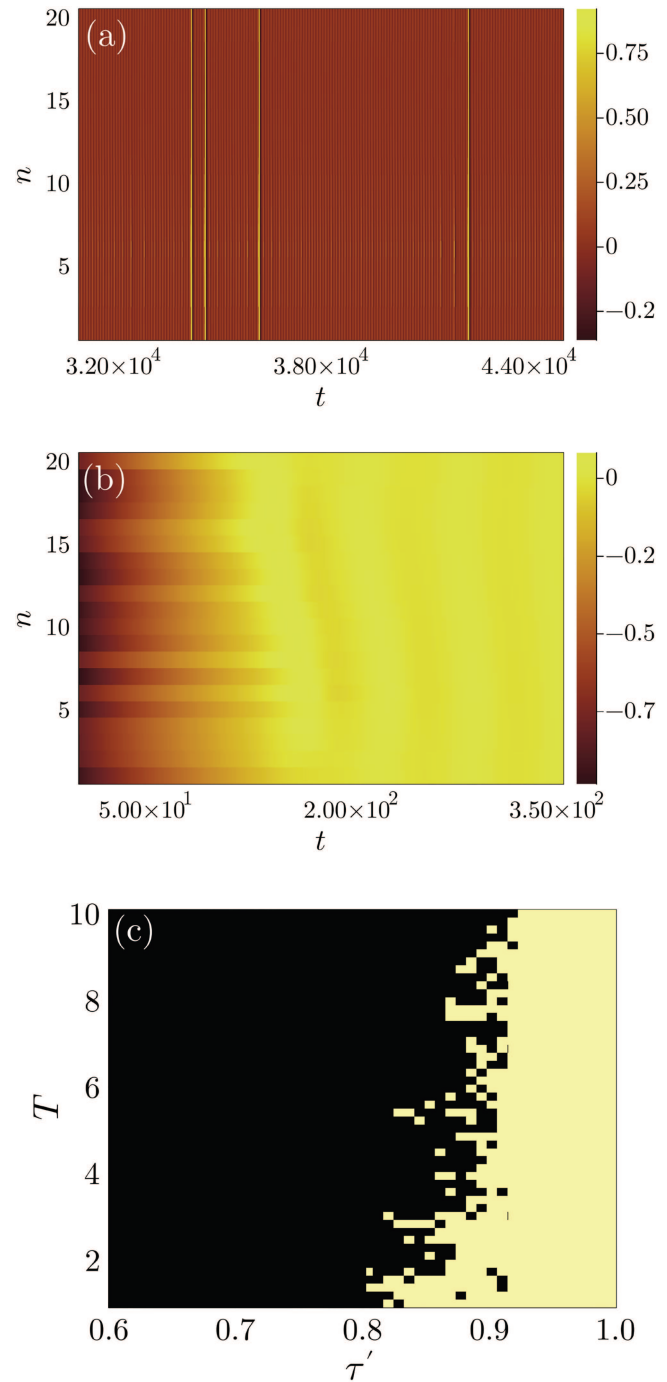


FIG. 13. Small-world network of FHN model with eight nearest neighbors and 0.1 rewiring probability: (a) temporal evolution of extreme events appeared for the coupling parameter $k = 0.761$, $T = 1.0$, and $\tau' = 1.0$ and (b) quenching of oscillation for $k = 0.761$, $T = 1.0$, and $\tau' = 0.5$. (c) Phase diagram in the τ' vs T plane for a fixed k value = 0.761 manifest mitigation of extreme events in a specific range of time-varying parameter values.

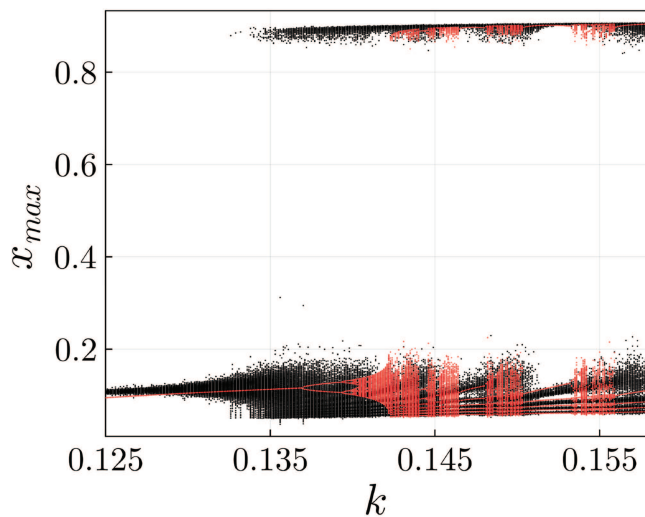


FIG. 14. Bifurcation diagram depicting the influence of small fluctuation in the on and off states of the coupling: (a) $k_c = 0.132$ with $\tau' = 0.9$ (black) while including a small amount of Gaussian noise in the coupling function with the amplitude of 10^{-4} .

network of FHN units. The small-world network is established using the connectivity matrix A_{ij} , with each system connected to its eight nearest neighbors. The rewiring probability is set at 0.1. The other system parameters are fixed as $a = -0.02651$, $c = 0.02$, and $k = 0.761$. For time-varying parameter of $T = 1.0$ and $\tau' = 1.0$ [cf. Fig. 13(a)], the system manifests rare extreme events. On the other hand, for $T = 1.0$ and $\tau' = 0.5$, the FHN units exhibit only amplitude death which is depicted in Fig. 13(b), which denotes the extreme events are completely eliminated from the system. To ascertain the mitigation region of extreme events, we have plotted the phase diagram in a specific range of duty cycle percentage $\tau' \in (0.6, 1.0)$ and interaction time $T \in (1, 10)$ as shown in Fig. 13(c). It can be obtained from Fig. 13(c), regardless of the network configuration, we have identified the possibility of controlling the unexpected extreme events in a network of FHN units.

V. CONCLUSION

We have investigated the influence of time-varying interaction in the generation and annihilation of extreme events in two different model systems, namely, the FHN neuron model and the forced Liénard system. Our study revealed the sporadic occurrence of unexpected extreme events across broader parameter ranges, influenced by the interaction time between these systems. The results show that the interaction duration, period of interaction time, and coupling strength play a vital role in determining the appearance of extreme events or its annihilation. The emergence of extreme events was confirmed using a significant height threshold measure and probability distribution function. Further, we have identified that with an appropriate choice of time-varying parameters in coupled oscillators, the unwanted rare large-amplitude events can control based on the specific requirements. Noteworthy, we proved that the

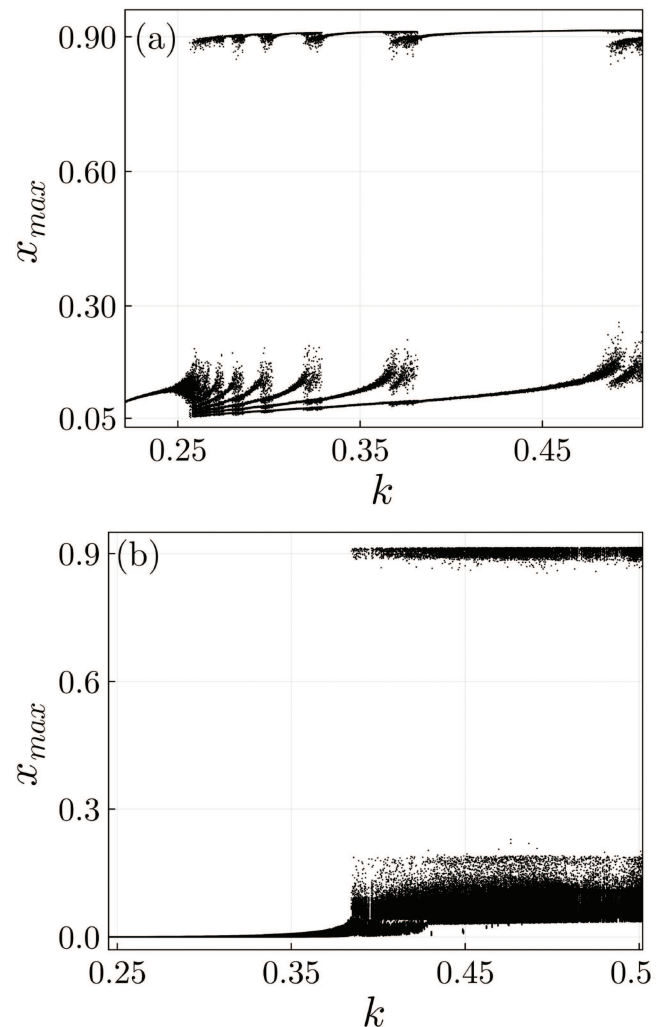


FIG. 15. The bifurcation diagram of the coupled FHN model with a stochastically varying coupling function illustrates the system's transition from chaos to extreme events for a specific coupling strength, denoted as k . In (a) and (b), we examine two different time periods, $T = 1.0$ and $T = 10$, respectively. For the time period $T = 1.0$, the transition occurs at $k = 0.254$, while for $T = 10$, the critical value of the transition is $k = 0.386$. During the time period T , the on and off states of the coupling change randomly.

time-varying interaction induced the appearance and mitigation of extreme events and amplitude death dynamics in distinct networks of the neuron model.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

S. Leo Kingston: Conceptualization (equal); Formal analysis (equal); Funding acquisition (equal); Investigation (equal); Methodology (equal); Validation (equal); Visualization (equal); Writing – original draft (equal); Writing – review & editing (equal). **Gayathri Kumaran:** Data curation (equal); Formal analysis (equal); Investigation (equal); Validation (equal). **Anupam Ghosh:** Investigation (equal); Validation (equal); Visualization (equal); Writing – original draft (equal); Writing – review & editing (equal). **Suresh Kumarasamy:** Conceptualization (equal); Formal analysis (equal); Investigation (equal); Methodology (equal); Supervision (equal); Writing – review & editing (equal). **Tomasz Kapitaniak:** Conceptualization (equal); Funding acquisition (equal); Project administration (equal); Supervision (equal); Writing – original draft (equal); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

APPENDIX: EFFECT OF STOCHASTIC TIME-VARYING INTERACTION

In the main text, we discussed the dynamics of coupled systems and networks of systems under periodically varying interactions. In this Appendix, we delve into the dynamics of the coupled system under non-periodic variations in coupling between the systems. First, we consider the impact of small random variations during on-off interactions. In the intervals $0 < t \leq \tau$, α exhibits random small-amplitude fluctuations, while during $\tau < t \leq T$, β undergoes similar small variations.

For this analysis, we fixed the system parameters as $a = -0.025794$, $b_1 = 0.0065$, $b_2 = 0.0135$, $c = 0.02$, $\tau' = 0.9$, and $T = 1.0$ and varied the coupling strength k within the range (0.125, 0.16), including small random fluctuations in the on and off states of the time-varying interaction. The obtained results are plotted as a bifurcation diagram in Fig. 14, revealing that due to the small instability (with the Gaussian noise with amplitude of 10^{-4}) during the interaction time, we observed a significant change in the transition points. To illustrate this, we overlapped the resultant plot (red-colored bifurcation diagram) with the bifurcation diagram without random fluctuation in Fig. 2(b). Furthermore, it is evident from Fig. 14 that small random variations lead to a smooth transition in the coupled FHN model.

Next, we investigate the effects of entirely stochastic interactions, where links between the FHN units vary stochastically over time (non-periodically). During the time period T , the on and off states of the coupling undergo random changes, indicating that the

frequencies of on and off states are not the same. We used the same system parameters as mentioned at the beginning of the Appendix, and the bifurcation diagram for two different interaction times is portrayed in Figs. 15(a) and 15(b). In Figs. 15(a) and 15(b), we examine two different time periods, $T = 1.0$ and $T = 10$, respectively. For the time period $T = 1.0$, the transition occurs at $k = 0.254$, while for $T = 10$, the critical value of the transition is $k = 0.386$. Once again, we identified significant changes in the transition points as well as bifurcation structures. Thus, exploring distinct periodic and stochastic interactions induced extreme events in unexpected regions helps to understand the overall possibilities of the occurrence of extreme events in a model system.

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